

Phylogeny Construction

1. Consider the following distance matrix on four taxa.

	a	b	c	d
a	0	4	5	4
b		0	3	4
c			0	5
d				0

Apply an additive phylogeny algorithm and find the tree.

2. Use the same matrix in the previous question and apply the UPGMA algorithm to find the tree.
3. What are the possible applications of a phylogenetic tree? How related are hierarchical clustering algorithms and phylogeny construction algorithms?
4. What is an ultrametric tree? How is that relevant in evolutionary understanding?
5. Prove that any tree with n leaves and no degree-2 internal nodes has exactly $2n-2$ total nodes.
6. Show that in a tree, the sum of degrees over all nodes equals $2(n-1)$, where n is the number of nodes.
7. Given the following matrix:

0 8 14 14
 8 0 14 14
 14 14 0 8
 14 14 8 0

 Verify whether it satisfies the four-point condition.

8. Given:

$D = \begin{pmatrix} 0 & 5 & 9 & 9 & 8 \\ 5 & 0 & 10 & 10 & 9 \\ 9 & 10 & 0 & 8 & 7 \\ 9 & 10 & 8 & 0 & 3 \\ 8 & 9 & 7 & 3 & 0 \end{pmatrix}$

Compute the D^* for Neighbor Joining. Identify the pair chosen for joining. Compute the branch lengths to the new internal node.

9. Apply the Additive Phylogeny algorithm to:

$\begin{pmatrix} 0 & 7 & 11 & 14 \\ 7 & 0 & 10 & 13 \end{pmatrix}$

11 10 0 7
14 13 7 0)

Identify a leaf with minimum limb length. Remove it and construct the reduced matrix. Reinsert the leaf and show where it attaches.

10. Explain how small parsimony works with dynamic programming.

Hidden Markov Models

1. What is the log-odds ratio? How can it be used to find states from observations in HMM? What are the problems with using log-odds ratio to decode HMM states?
2. We have a 2-state HMM (Background B, Island I). Initial probabilities: $P(B) = 0.5$, $P(I) = 0.5$. Transition probabilities: $P(B \rightarrow B) = 0.85$, $P(B \rightarrow I) = 0.15$; $P(I \rightarrow I) = 0.85$, $P(I \rightarrow B) = 0.15$. Emission probabilities:
 $E_B : P(A) = 0.25, P(C) = 0.25, P(G) = 0.25, P(T) = 0.25$
 $E_I : P(A) = 0.1, P(C) = 0.4, P(G) = 0.4, P(T) = 0.1$
Sequence: $x = G C T$
Fill the Viterbi DP table and report the most likely state path.
3. Suppose you have the following sequence
ACCGCCGT
Considering all distributions uniform, run 2 iterations of the Baum-Welch Algorithm (use pseudo-count = +1).
4. What is the runtime complexity of the viterbi algorithm?
5. Can Baum-Welch algorithm guarantee optimal model and parameters? What is the solution to this problem?
6. Viterbi algorithm may result in underflow. What is the solution to this problem?
7. What is the space complexity of the Viterbi algorithm? Explain.

Clustering Gene Expression Data

1. Consider these data points, (1,2), (5,6), (4,3) and (6,4). Now initialize any two of them as cluster centers for a clustering with $k=2$. Now iterate 1 step for K-Means and Soft K-Means clustering (using $\beta=0.25$)
2. For the previous data, if initially the first point is selected as a cluster center, what are the probabilities of the other points being selected as the second center?
3. What are the limitations of a Hard K-Means Clustering Algorithm? How can each of these be mitigated or handled to improve the performance of the algorithm?
4. Why is clustering genes important in Biology?
5. What are the biological motivations for Soft-Kmeans clustering and hierarchical clustering?

6. You are given expression values for 4 genes across 4 experimental conditions:

GeneA: [2.0, 3.0, 2.5, 3.5]

GeneB: [2.2, 3.1, 2.7, 3.6]

GeneC: [5.0, 5.5, 5.1, 5.4]

GeneD: [8.0, 7.5, 7.8, 7.7]

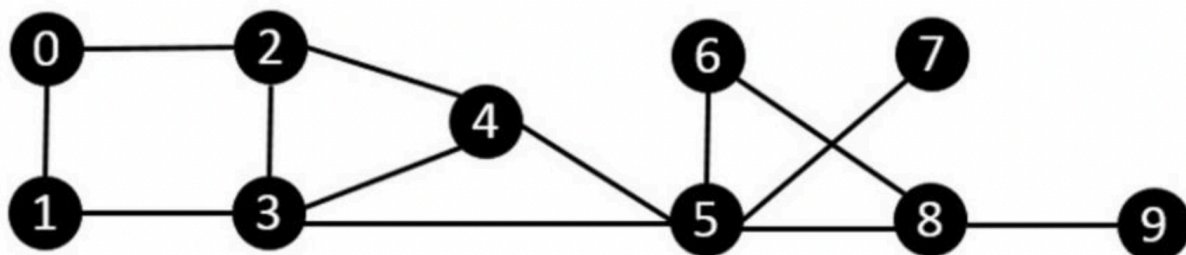
Perform Agglomerative Hierarchical Clustering using, distance metric: Euclidean distance, linkage method: single linkage (minimum distance between any two points in clusters)
Approach: Bottom-up (merge closest clusters).

7. Repeat the previous problem using average distances.
8. What are the possible disadvantages of hierarchical clustering?

Dimensionality Reduction

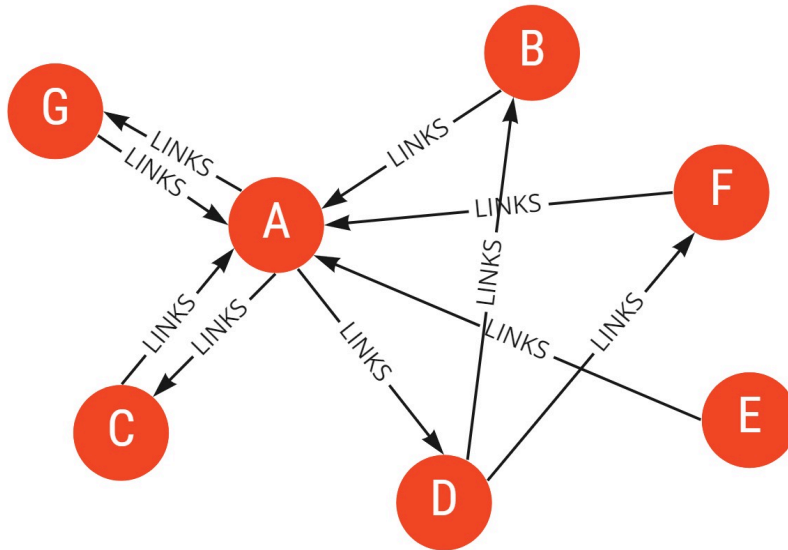
1. Why do we need dimensionality reduction for gene expression data?
2. Among three methods shown in class, PCA, MDS and SNE, which do you prefer and why? Explain pros and cons of each method. What do each of these methods try to minimize?
3. What is KL-divergence? Explain how SNE works with an example of input space of two dimensions and output space of one dimension. Illustrate using 3 sample data points: (2,1), (3,4), (0,3).
4. Explain MDS and PCA with the same data points. Explain the working principles only.
5. Why is distance based metric like euclidean distance / L1 metric/L3 metric not suitable for SNE?
6. Explain Soft Neighborhood. Why is the squared distance used in the soft-neighborhood formula?
7. You want to reduce dimensionality using PCA. How do you select the output dimensionality?
8. In gene expression data, suppose you are using PCA to reduce dimensions, is it possible to know which input dimensions or genes are important using PCA?

Network Analysis



1. For the above graph perform the following:

- a. Find the diameter
 - b. Find closeness centrality of node 4
 - c. Find betweenness centrality of node 4 and edge between 4-5
 - d. Rank the nodes and edges (use any of the centrality measures)
2. Consider the following network:



Now simulate random walk on this network to rank the nodes. Go up to 2 iterations. Initialize the probability distribution as 1/7 initially and after each iteration, show the updated values.

3. Provide examples of directed , undirected and weighted networks in Biology.
4. What are the types of analysis that are done on Biological Networks?
5. What are the advantages and disadvantages of a scale-free network?
6. Explain the small world effect and its implications for Protein-Protein Interaction Networks.